

# Pinzhang Cheng (Jack)

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## EDUCATION

### University of New South Wales

Sep. 2024 – May. 2026

*Master of Mathematics | WAM: 86.67*

Sydney, Australia

- **Coursework:** Optimization, Data & Machine Learning, Applied Linear Algebra, Stochastic DEs, Stochastic Analysis, Discrete/Continuous Time Financial Modelling, Computational Mathematics for Finance, Functional Analysis

### University of Sydney

Feb. 2024 – Jun. 2024

*Science Undergraduate Non-Degree | WAM: 86.5*

Sydney, Australia

- **Coursework:** Probability & Statistical Models (Adv), Linear and Abstract Algebra (Adv), Nonlinear ODEs (Adv), Vector Calculus and Differential Eqs (Adv)

*Bachelor of Economics (Honours Class I) | WAM: 80.59*

Feb. 2020 – Nov. 2023

- **Coursework:** Game Theory, Microeconomics, Macroeconomics, Econometrics, Metric Spaces (Adv), Lagrangian and Hamiltonian Dynamics, Derivative Securities

## EXPERIENCE

### University of Sydney

Feb. 2024 – Apr. 2024

*Sessional Tutor*

Sydney, Australia

- Tutored around 20 undergraduate students in Quantitative Methods in Economics, covering differentiation, optimization, multivariable functions, Linear Algebra; received 5 requests to serve as academic referee

## PROJECTS

### Master's Thesis: A Novel Algorithm For Sparse Portfolio Selection

Jul. 2025 – Apr. 2026

- Developed DASH, a novel algorithm for large-scale sparse portfolio selection formulated as convex MIQP; established theoretical properties and validated design choices through numerical simulations
- Identified key problem characteristics contributing to the computational difficulty, measured by the solve time and solution quality of commercial solver Gurobi
- Demonstrated consistent early-stage solution quality improvement (30%-250% depending on the configuration) over Gurobi in difficult problems with fixed time budgets (1-30 minutes)
- Performed sensitivity analysis to show the robustness of improvement and the expansion of improvement margin for problems with more difficult properties
- Generalised DASH as a metaheuristic for any subset selection problem that can be formulated in the same framework such as sparse feature learning in linear regression without regularization

### Honours Thesis: Zero Lower Bound in a Two-Asset HANK Macroeconomic Model

Jan. 2023 – Nov. 2023

- Built and solved a two-asset HANK model with dynamic programming in MATLAB; avoided the curse of dimensionality with an empirical portfolio rule and introduced real wage rigidity as the key friction driving inflation dynamics, backed by a formal necessity proof
- Calibrated model parameters to match Australian unemployment rate and wealth distribution data (ABS)
- Simulated impulse response functions of the model under supply and demand shocks and analyzed various propagating channels and agent behaviours

## SKILLS

**Languages:** Python (NumPy, SciPy, Pandas, CVXPY), MATLAB

**Tools:** LaTeX, Gurobi